

Multiple Choice

1. **Answer:** A

Options: A) inductance.; B) resistance.; C) impedance.; D) voltage.; E) capacitance.

Note: Correct option: A - inductance.

2. **Answer:** A

Options: A) 75%; B) 86%; C) 95%; D) 60%; E) 80%

Note: Correct option: A - 75%

3. **Answer:** C

Options: A) $(1/2)(e^{2t} - 1)$; B) $(1/2)(1 + \cos 2t)$; C) $(1/4)(1 - \cos 2t)$; D) $(1/4)(\sin(2t) - \cos(2t))$; E) $(1/2)\sin(2t)$

Note: Correct option: C - $(1/4)(1 - \cos 2t)$

4. **Answer:** B

Options: A) $0.1 \cos 10^6 t$; B) $100 \cos 10^6 t$; C) $1 \cos 10^6 t$; D) $10^{-3} \cos 10^6 t$; E) $0.01 \sin 10^6 t$

Note: Correct option: B - $100 \cos 10^6 t$

5. **Answer:** A

Options: A) $t \sin t, (1/\sqrt{\pi}) \text{ ullet } (1/)$; B) $\cos t, (1/\sqrt{\pi}) \cdot t e^{\{-t^2\}}$; C) $e^t \sin t, (1/2\sqrt{\pi}) \cdot t$; D) $t \cos t, (1/\sqrt{\pi}) \text{ ullet } (1/)$; E) $t^2 \sin t, (1/2\sqrt{\pi}) \cdot (1/)$

Note: Correct option: A - $t \sin t, (1/\sqrt{\pi}) \text{ ullet } (1/)$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'tetravalant'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'LSB, Least Significant Bit'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.004 A'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $1 - e^{-(x+y+z)}$ for $x, y, z > 0$. Explain why this is the correct answer and provide supporting evidence.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

11. **Answer:** Multiplexers. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key